26900201 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECECT302 - LOW VOLTAGE SYSTEM FOR BUILDING MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. HVAC is an important BMS function because it [U] / [CO1]

   a). Consumes a significant portion of building energy

   b). Controls only security

   c). Controls elevators only

   d). Does not use energy

2. During a fire, BMS can automatically [U] / [CO1]

   a). Activate alarms

   b). Shut down air handling systems

   c). Control smoke extraction

   d). All of the above

3. BMS contributes to [U] / [CO1]

a). Environmental sustainability

b). Energy wastage

c). Higher maintenance costs

d). Poor comfort

4. The main purpose of a BMS is to [U] / [CO1]

a). Monitor, manage, and optimize building services

b). Only control lighting

c). Only control elevators

d). Only monitor security

5. Daylight harvesting can [U] / [CO1]

a). Increase unnecessary lighting

b). Increase energy consumption

c). Reduce artificial lighting

d). Disable sensors

6. Which level contains sensors and actuators? [U] / [CO1]

a). Field Level

b). Automation Level

c). Management Level

d). Cloud Level

7. The communication network connects controllers with [U] / [CO1]

a). Servers and other BMS devices

b). Building walls

c). Fire extinguishers

d). None of these

8. Which is a critical alarm example? [U] / [CO1]

a). Chiller failure

b). Normal temperature

c). Normal lighting

d). Occupied room

9. . BMS can monitor elevator [U] / [CO1]

a). Status

b). Traffic patterns

c). Car positions

d). All of these

10. A BMS can reduce lighting energy by [U] / [CO1]

a). Turning off unnecessary lights

b). Turning all lights on continuously

c). Removing occupancy sensors

d). Increasing brightness permanently

11. The main purpose of voice evacuation is to [U] / [CO1]

a). Provide clear emergency instructions

b). Monitor energy

c). Control pumps

d). Detect smoke

12. PIN codes are used to [U] / [CO1]

a). Measure temperature

b). Authenticate users

c). Detect smoke

d). Control pumps

13. Unlike rheostat dimming, TRIAC–DIAC dimming controls brightness by [A] / [CO2]

a). Dissipating excess power as heat

b). Controlling the conduction period of the AC waveform

c). Increasing the resistance of the lamp

d). Reducing the supply frequency

14. Thermocouples are particularly suitable for [A] / [CO2]

a). Measuring a wide range of temperatures

b). Measuring humidity only

c). Detecting occupancy

d). Measuring air pressure only

15. Lighting levels can be adjusted according to [A] / [CO2]

a). User requirements

b). Environmental conditions

c). Both A and B

d). Neither A nor B

16. What is a major advantage of solid-state relays compared with electromagnetic relays? [A] / [CO2]

a). They contain more moving parts

b). They provide faster switching, silent operation, and longer service life

c). They require mechanical springs

d). They have lower reliability

17. Which HVAC faults can airflow measurements help detect? [A] / [CO2]

a). Blocked ducts

b). Malfunctioning fans

c). Both A and B

d). Neither A nor B

18. High-frequency PWM switching can generate [A] / [CO2]

a). Mechanical vibration only

b). Water pressure

c). Natural daylight

d). Electromagnetic interference (EMI)

19. Inverter-driven compressors help reduce [A] / [CO2]

a). Refrigerant flow

b). Building occupancy

c). Airflow

d). Temperature fluctuations

20. In an HVAC system, if room temperature rises above the desired setpoint, the controller may command: [A] / [CO2]

a). Valve actuators

b). Damper actuators

c). Fan speed controllers

d). All of the above

21. What is the main function of the heating subsystem? [A] / [CO2]

a). Remove excess heat

b). Generate heat during cold weather

c). Remove humidity only

d). Supply chilled water

22. By adjusting valve positions, HVAC systems can precisely control [A] / [CO2]

a). Heat transfer

b). Building lighting

c). Occupancy

d). Electrical frequency only

23. At minimum rheostat resistance, the lamp operates [A] / [CO2]

a). At or near its full rated brightness

b). At minimum brightness

c). Completely OFF

d). Only at half brightness

24. What do door sensors monitor? [A] / [CO2]

a). Elevator speed

b). Motor temperature only

c). Whether the elevator doors are open or closed

d). Cabin weight only

25. What is the main function of a VFD in a fan system? [A] / [CO2]

a). Adjust the fan motor speed

b). Measure room temperature

c). Control water flow directly

d). Detect smoke

26. Which of the following is an advantage or application of IR/PIR sensors? [A] / [CO2]

a). Low power consumption and fast response

b). Automatic doors and motion-activated lighting

c). Intrusion detection and occupancy monitoring

d). All of the above

27. Which of the following is a component of a typical lighting control system? [A] / [CO2]

a). Occupancy sensor

b). Daylight sensor

c). Controller

d). All of these

28. If the desired room temperature is 24°C and if deadband is used, when does the cooling system turn ON? [A] / [CO2]

a). Below 23°C

b). At exactly 24°C

c). Above 25°C

d). Above 30°C

29. PIR sensors are particularly useful for detecting [An] / [CO3]

a). Water leakage

b). Electrical voltage

c). Human movement

d). Door locking

30. What is the main purpose of an access card? [An] / [CO3]

a). Detect smoke

b). Control fan speed

c). Measure temperature

d). Identify or authenticate a user

31. What diagnostic capability is provided by an addressable fire alarm system [An] / [CO3]

a). It only activates bells

b). It controls building temperature

c). It replaces all sensors

d). It can provide detailed information about device status and faults

32. What can unusually high energy consumption during unoccupied hours indicate? [A] / [CO4]

a). Unnecessary operation of equipment

b). Perfect energy efficiency

c). Reduced equipment operation

d). Zero energy consumption

33. What is the main objective of a fire protection system? [A] / [CO4]

a). Reduce lighting levels

b). Detect, control, contain, and respond to fire

c). Increase energy consumption

d). Control elevator speed only

34. What is the main purpose of a Manual Call Point (MCP)? [A] / [CO4]

a). Automatically detect smoke

b). Allow a person to manually activate the fire alarm

c). Control building lighting

d). Detect temperature

35. Which of the following is a contaminant that ventilation can help reduce? [A] / [CO2]

a). Carbon dioxide

b). Dust particles

c). Volatile organic compounds (VOCs)

d). All of the above

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. What does CCTV stand for? Explain briefly. [U] / [CO1]

37. Discuss the advantages of TRIACs over mechanical relays. [A] / [CO2]

38. Discuss the role of IDS as a layer of physical security. [A] / [CO3]

39. How do controllers use VFDs to control pumps and fans? [U] / [CO1]

40. Discuss the advantages and applications of DCV in modern buildings. [A] / [CO2]

[5x3 = 15]